Year 8 Science – Booklet 26: Physics

Energy Resources (Using Energy Resources) — ANSWERS

> Note: this booklet's content is identical to Year 8 Booklet 9 (Energy Resources), so the same worked answers apply throughout.

Quick Test (p.109)

1. Coal, oil, and gas.
2. Carbon dioxide.
3. Coal and oil.
4. Any one of: air pollution/acid rain, global warming (greenhouse effect), environmental damage from mining, oil spillage during transport.
5. Because they take millions of years to form, so once used up they cannot be replaced within a useful timescale.
6. Any three of: reduce petrol use (smaller cars, public transport, walking/cycling), use more efficient car engines, improve home/factory insulation, raise awareness to reduce energy waste, develop renewable energy sources.

Section A – Multiple Choice (5 marks)

1. a) wood
2. b) carbon dioxide
3. b) coal
4. d) driving bigger cars
5. d) turn turbines

Section B (20 marks)

1. Fill in the missing words (13 marks)

Coal, oil and **gas** are called **fossil** fuels. They are concentrated **sources** of energy. They are formed from the remains of **plants** and **animals** that died millions of years ago. But instead of rotting they became covered with many layers of **mud**. As a result they were under great **pressure** and they were at high **temperature**. These conditions changed them into **fossil** fuels. Because these fuels take a long time to form they are called **non-renewable** fuels. Once they have been used up they cannot be **replaced**. Burning these fuels can cause problems to the environment such as **acid rain** and the **greenhouse effect**.

2. Anagrams and their meaning (7 marks)

a) sssfflloeui → **fossil fuels** – non-renewable energy resources (coal, oil, gas) formed from the remains of dead plants and animals over millions of years.
b) oottnaprewsi → **power stations** – places where fuel is burned to generate electrical energy, distributed via the National Grid.
c) iiaadrcn → **acid rain** – rainfall made acidic by pollutant gases from burning fossil fuels, which can damage buildings, trees and water habitats.
d) eeeeeftuoghncrs → **greenhouse effect** – the warming of the Earth's atmosphere caused by gases like carbon dioxide trapping heat that would otherwise escape into space.
e) oponilltu → **pollution** – the introduction of harmful substances into the environment, causing damage to it.

f) eeeanrlbw → **renewable** – describes an energy source (e.g. wind, solar, tidal) that won't run out, as it's naturally replenished.
g) nnneeebolawr → **non-renewable** – describes an energy source (e.g. coal, oil, gas) that will eventually run out, since it cannot be replaced once used up.

Section C (14 marks)

1. Power station diagram using natural gas

a). Chemical energy.

b). By burning (combusting) it.

c). To heat water and turn it into steam.

d). Heat energy (in the steam) is transferred into kinetic energy (turning the turbine).

e). Kinetic energy (of the turbine) is transferred into electrical energy (by the generator).

2. Natural gas is a fossil fuel

a). Coal and oil.

b). Because they take millions of years to form, so they cannot be replaced once used up.

c). Acid rain (from burning coal/oil) and the greenhouse effect/global warming (from CO■ emissions).

d). Improve insulation in homes/factories to waste less energy; use public transport or more efficient/smaller cars to reduce petrol consumption.

Note: This is a Collins "KS3 Science Success Guide" page-set ("Using energy resources"), reproduced here by Aristeia Academy Tuition, rather than an exam-board past paper – there is no publicly downloadable official markscheme for it. These answers were worked out directly from the notes and data given earlier in the booklet.